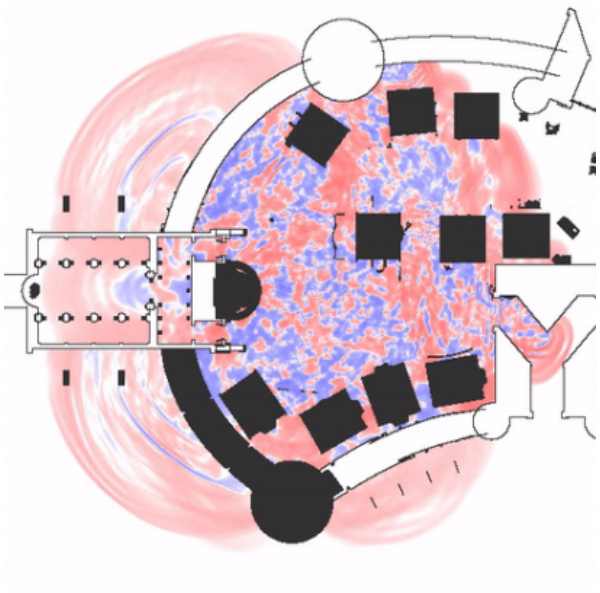
Fast algorithms and precomputation in room acoustics and planetary science

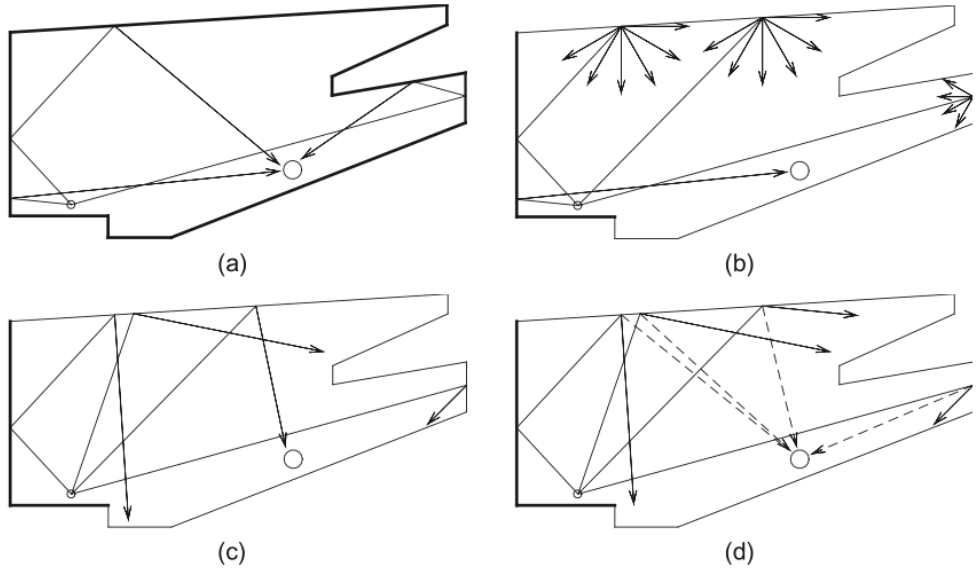
Samuel F. Potter

University of Maryland Dept. of Computer Science
https://umiacs.umd.edu/~sfp

July 22, 2021

State of the art solvers for room acoustics





Numerical acoustics: solve wave equation for bandlimited impulse input. High-frequency (>2kHz) simulations expensive and impractical.

Geometric acoustics: use raytracing to assemble room impulse response. Handling diffraction effects with raytracing is complicated.

Source: `https://www.microsoft.com/en-us/research/project/project-triton`

Geometric optics from Helmholtz

For sound speed c and freq. $\omega \gg 0$, the Helmholtz equation is:

$$\left(\Delta + \frac{\omega^2}{c^2}\right) U = 0$$

High-frequency geometric optics ansatz (WKB) is:

$$U(x) = \alpha(x) e^{i\omega\tau(x)} + O(\omega^{-1})$$

Here τ satisfies eikonal equation:

$$\|\nabla\tau\| = 1/c$$

and α is amplitude, which satisfies the transport equation:

$$\alpha\Delta\tau + 2\nabla\alpha^\top\nabla\tau = 0$$

Existing eikonal solvers don't work for our application

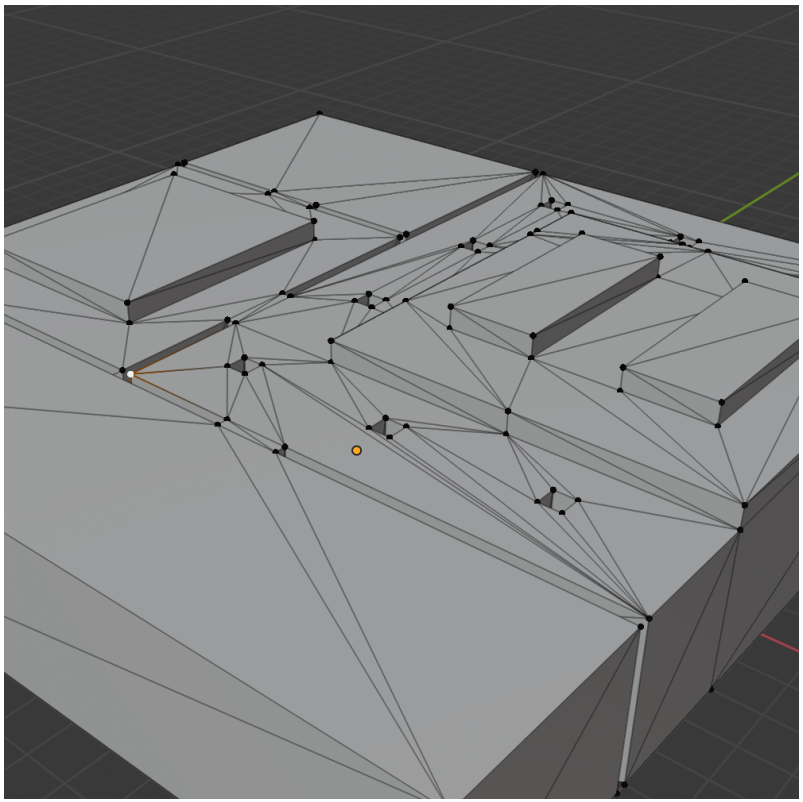
We need an eikonal solver that:

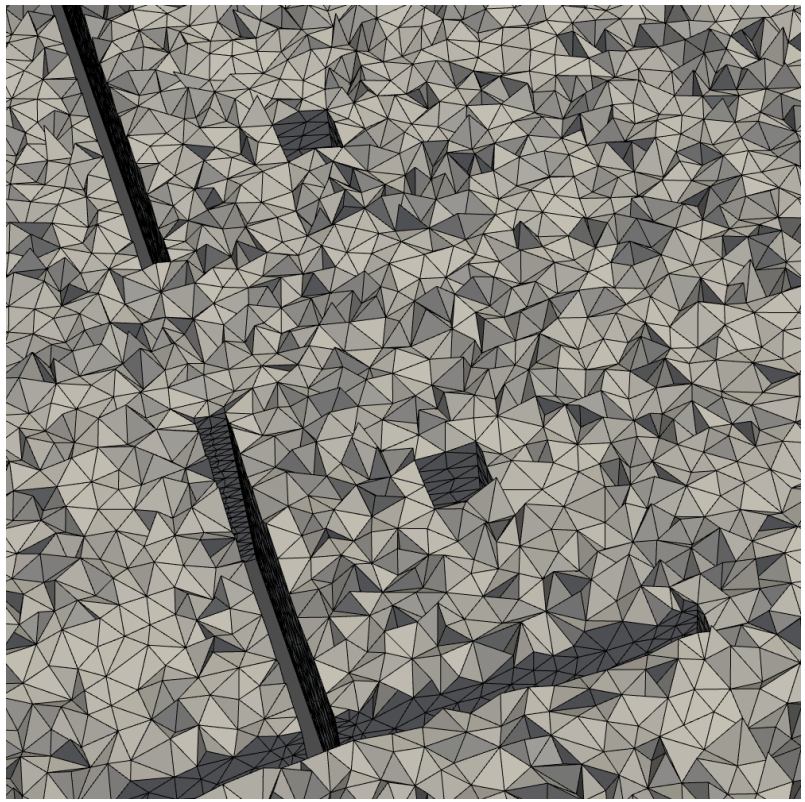
- Is at least $O(h^{3-p})$ accurate for $\partial^\alpha \tau$, $|\alpha| = p$
- Has a *compact stencil*
- Parametrizes local characteristics
- Is causal/is a direct solver
- Runs in optimal time ($O(N)$ or $O(N \log N)$ time)

Key idea: **design solver based on Hermite interpolation, semi-Lagrangian updates, and cell marching**

Numerical geometric acoustics (work in progress)

Develop JMM for solving the eikonal equation on a tetrahedron mesh and use this solver to compute acoustic parameters



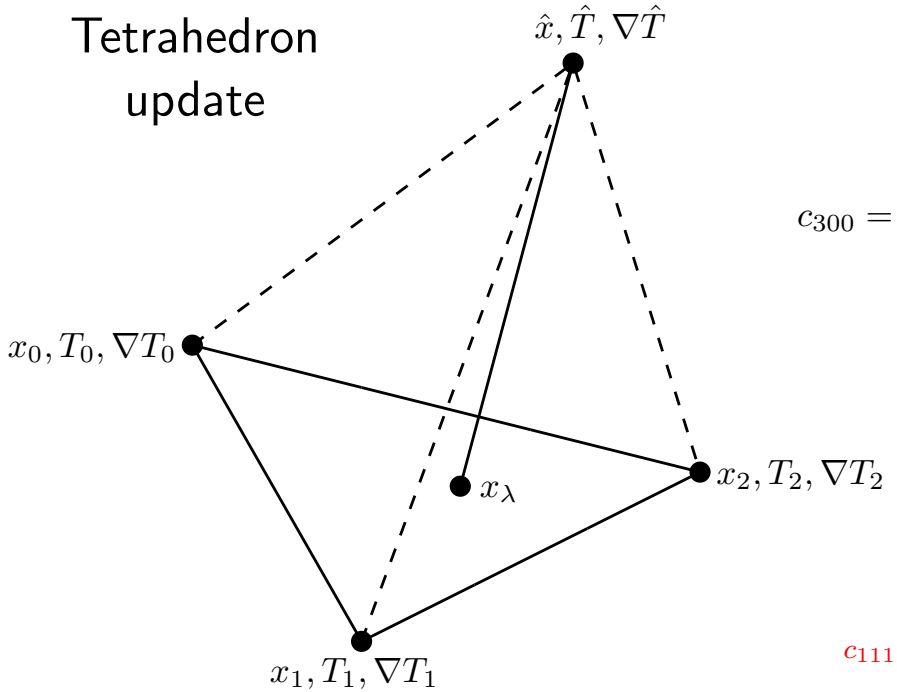


(A test scene made with Blender—mesh made using TetGen.)

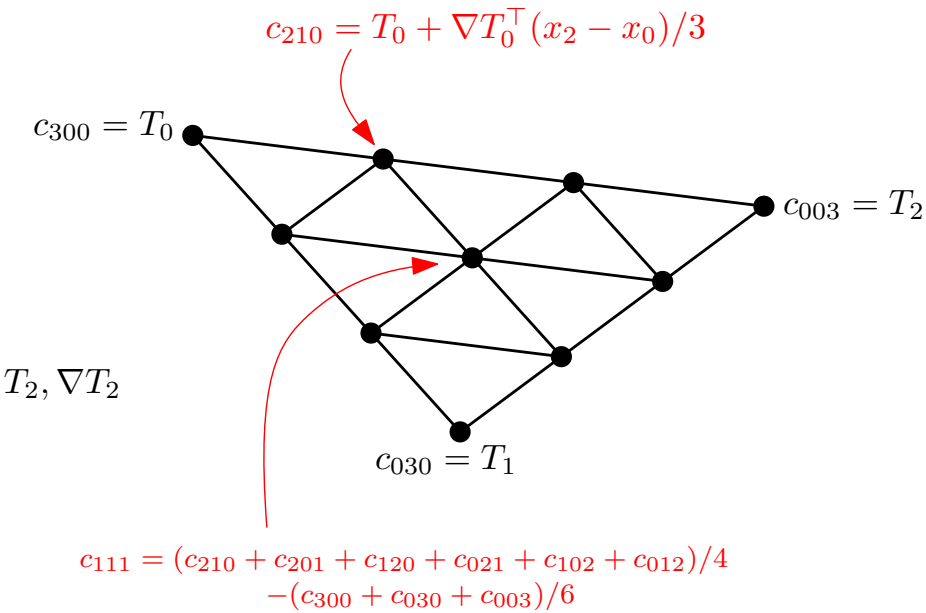
"Precomputed eikonal-based sound propagation", Potter, Cameron, and Duraiswami, *in preparation*.

Extension to 3D: initial focus on $c \equiv 1$

Tetrahedron update



Bernstein-Bézier triangle
($O(h^3)$ accurate)

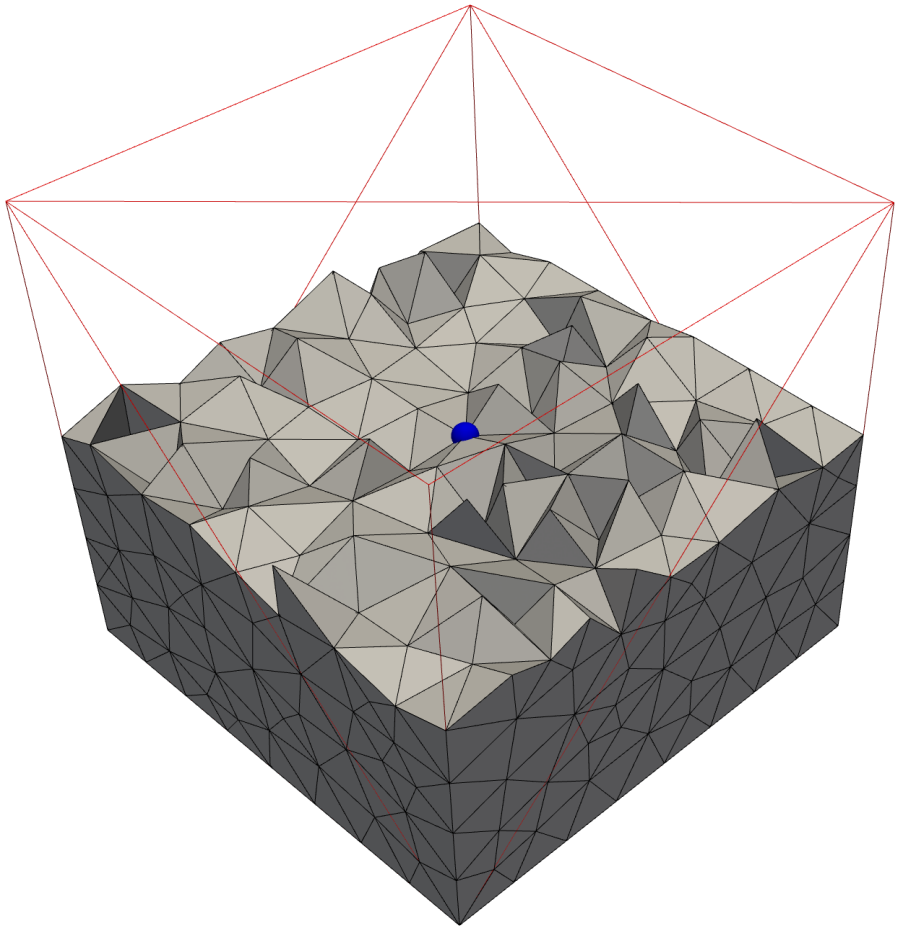


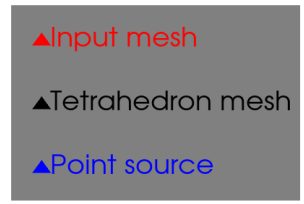
$$\hat{T} = \min_{\substack{\lambda_1 \geq 0, \lambda_2 \geq 0 \\ \lambda_1 + \lambda_2 \leq 1}} \left\{ T(x_\lambda) + \|\hat{x} - x_\lambda\| \right\}, \qquad \nabla\hat{T} = \frac{\hat{x} - x_{\lambda^*}}{\|\hat{x} - x_{\lambda^*}\|}$$

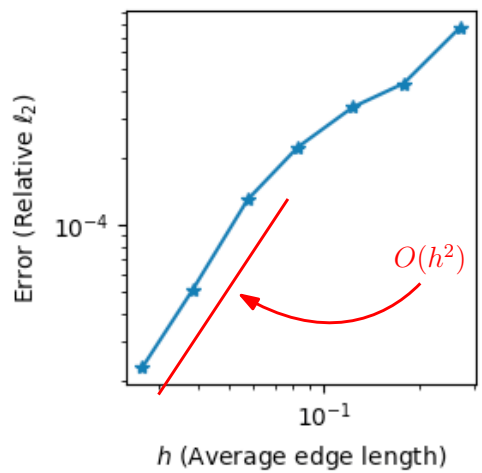
Accept update **only if** Lagrange multipliers are zero

"Precomputed eikonal-based sound propagation", Potter, Cameron, and Duraiswami, *in preparation*.

Extension to 3D: box domain







Domain is $\Omega = [-1, 1]^3$, place a point source at the origin so that $\tau(0) = 0$, use local factoring. Exact solution is simple: $\tau(x) = \|x\|$.

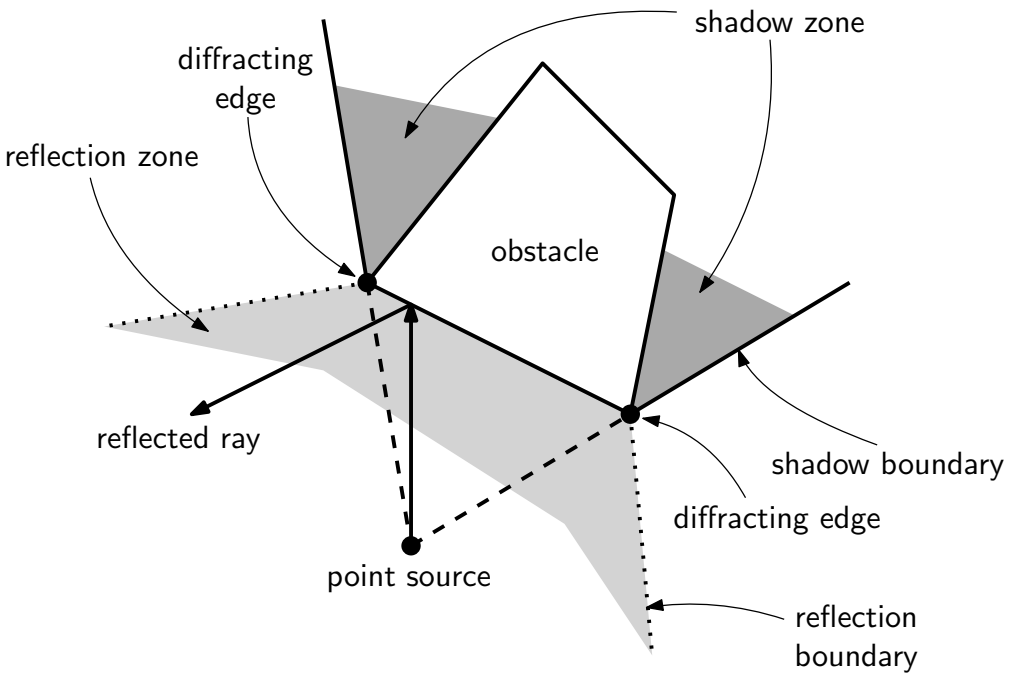
"Precomputed eikonal-based sound propagation", Potter, Cameron, and Duraiswami, *in preparation*.

Test problem: small building with a few rooms

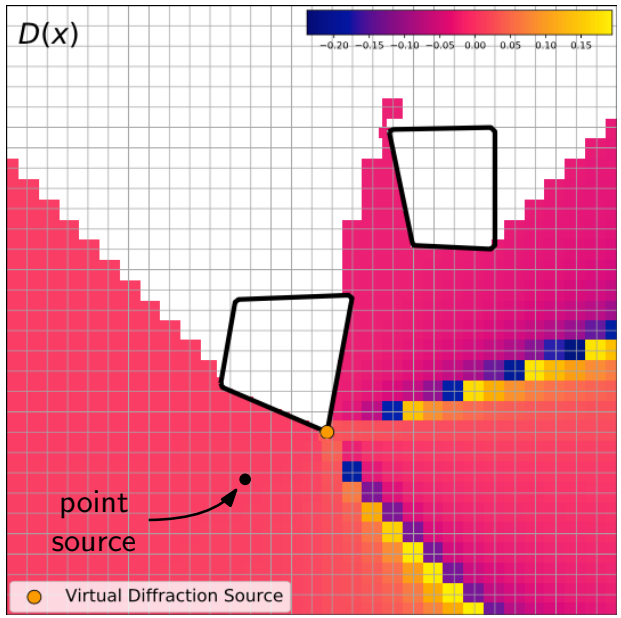
play movie

Numerical geometric acoustics: next steps

High-frequency diffraction



Computing the UTD coefficient in a 2D prototype

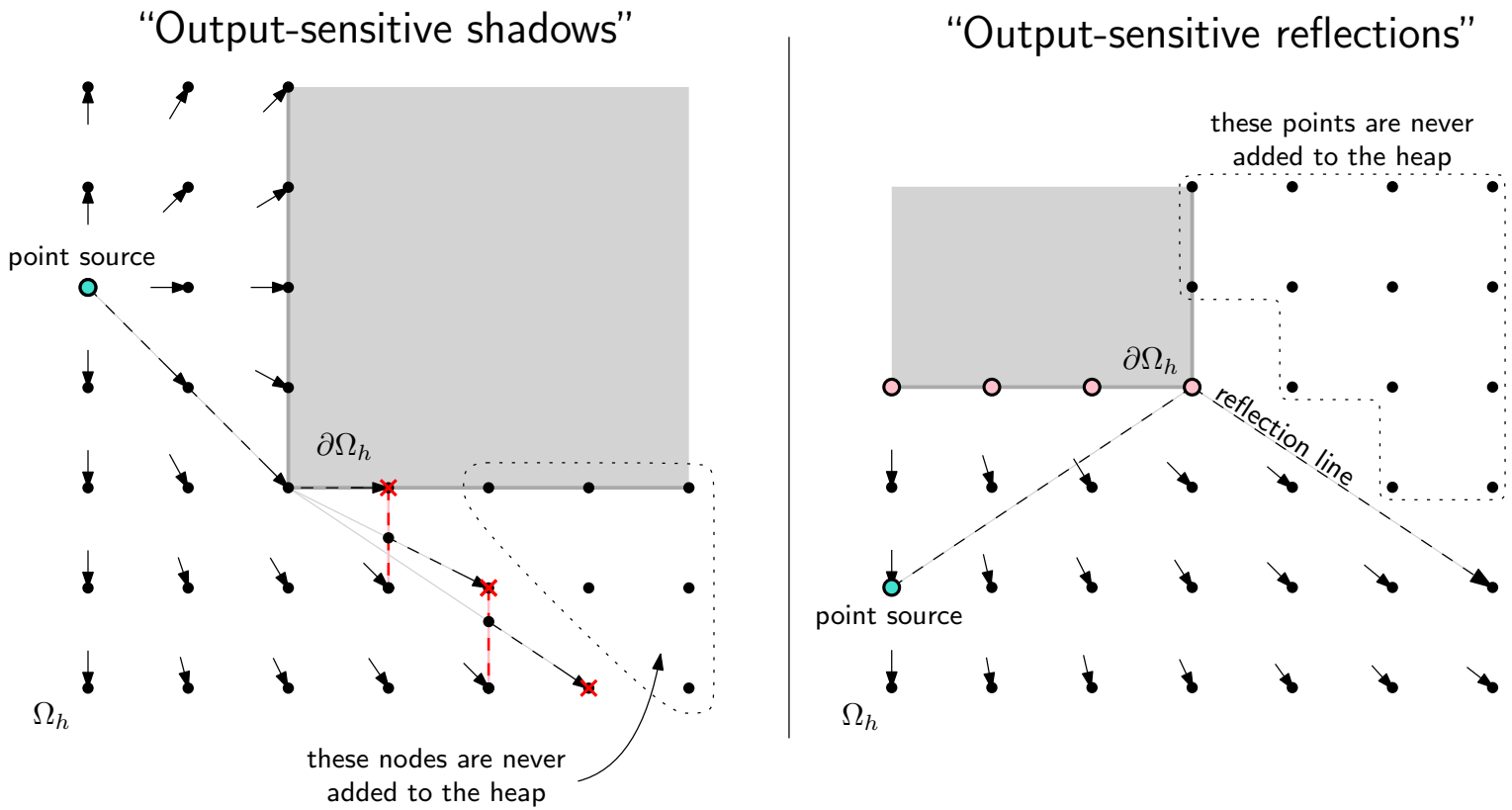


Developed algorithms to solve tree of eikonal problems to compute multiple arrivals due to reflection and diffraction in 2D.

Current work: *extend this to 3D.* Relies on the geometric/uniform theory of diffraction, and dynamic programming to march auxiliary variables.

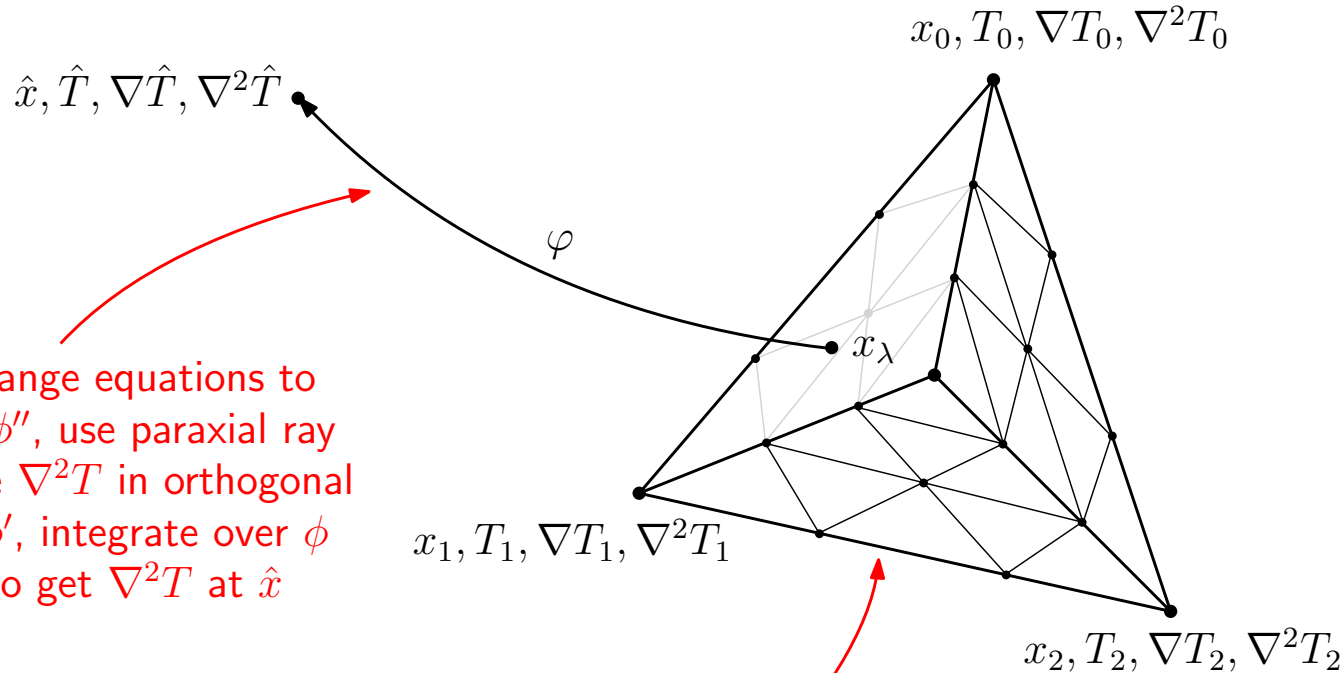
"Precomputed eikonal-based sound propagation", Potter, Cameron, and Duraiswami, *in preparation*.

Future research: output-sensitive JMMs



Idea: compute the "shadow cut" or "reflection cut" (nodes in refl. or shadow) and compute T *only* for directly visible nodes in $O(K)$ time, where $K =$ number of visible nodes

Future research: higher-order JMMs



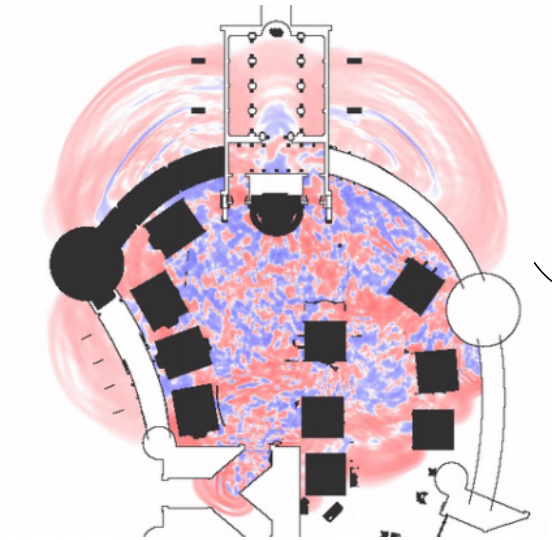
solve Euler-Lagrange equations to compute ϕ' and ϕ'', use paraxial ray theory to compute $\nabla^2 T$ in orthogonal complement of ϕ', integrate over ϕ and combine to get $\nabla^2 T$ at $\hat{x}$

e.g., use Clough-Tocher split for Hermite interpolate—Clough-Tocher splits exist that solve each higher-order Hermite interpolation problem†

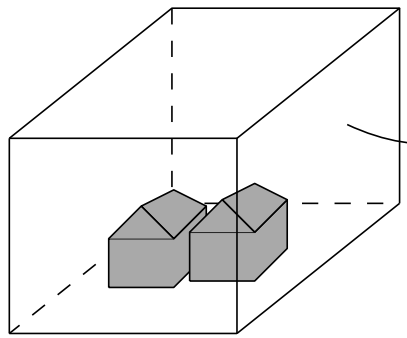
† Source: Lai, M.J. and Schumaker, L.L., 2007. *Spline functions on triangulations* (No. 110). Cambridge University Press.

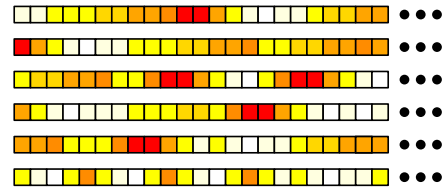
Spatial audio codecs for precomputed sound propagation

extract acoustic parameters from simulation of bandlimited impulse



downsampled volumetric acoustic parameter fields (single or multi-channel)





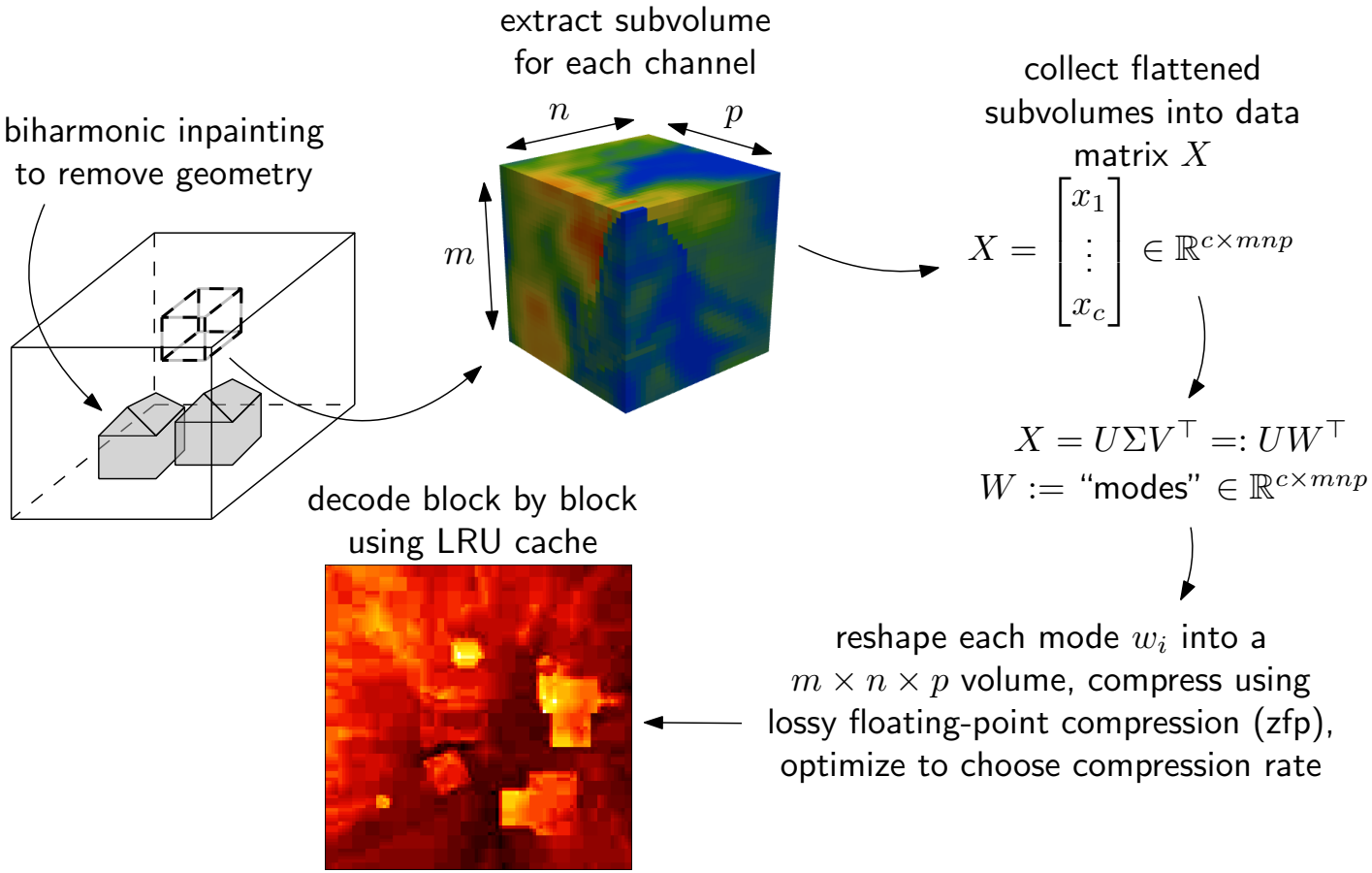
quantize using just noticeable differences (JNDs) and flatten

compute running differences and compress using zlib

Challenge:

- Many (42) parameters used to encode spatial audio
- Some are physically related to each other, others are multichannel parameters
- **Problem:** exploit relationships to reduce compressed size without compromising decode time

A block codec using lossy floating-point compression



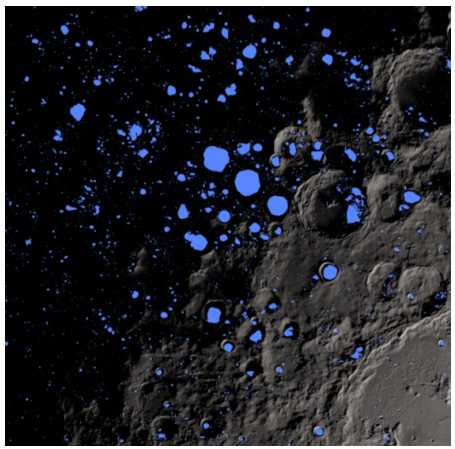
Results:

- Compression rate is **2x** better, for same decode speed
- Can take derivatives of decompressed fields (e.g., "first arrival time" = eikonal, "source directivity" = gradient of eikonal)

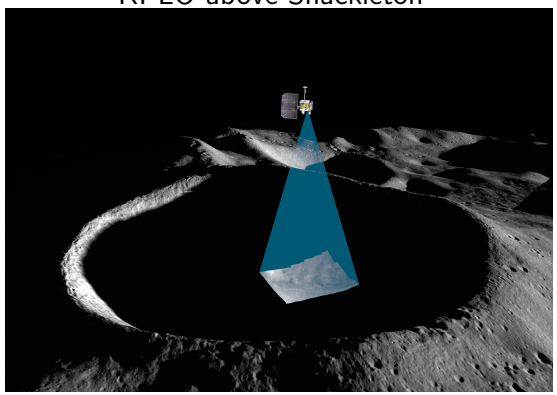
Thermal modeling in planetary science

Joint work with Erwan Mazarico (NASA), Norbert Schörghofer (PSI), and Stefano Bertone (NASA).

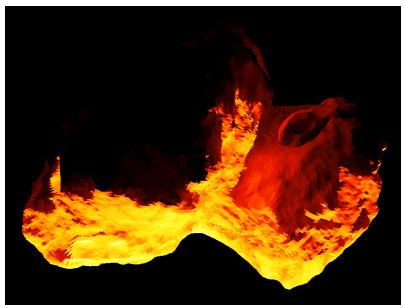
NASA SVS: "LRO Peers Into Permanent Shadows"†



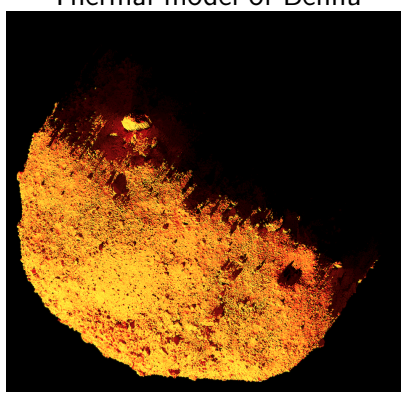
NASA ShadowCam on KPLO above Shackleton§



Thermal model of 67P



Thermal model of Bennu



- **Goal**: develop fast algorithms for Lambertian scattering
- Solve long-running thermal simulations (forward problem)
- Ultimate application is in inverse problems
- Target application is the study of Lunar PSRs

The NASA LROC imaged Lunar PSRs at approx. 20 m/px.
ShadowCam will image PSRs at approx. 1.7 m/px.

† Source: `https://svs.gsfc.nasa.gov/4043`.

§ Source: `https://www.nasa.gov/feature/nasa-selects-shadowcam-to-fly-on-korea-pathfinder-lunar-orbiter`

A thermal model on an airless body

Governing equations:

$$\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\left(k \frac{\partial T}{\partial z}\right) \qquad \text{(subsurface heat cond.)}$$

$$\epsilon \sigma T^4 = (1-\alpha)(E+B) + k\frac{\partial T}{\partial z} + \epsilon Q \qquad \text{(surface energy balance)}$$

$$B(x) = \frac{1}{\pi}\int_S \alpha(y)\Big(E(y)+B(y)\Big) f(x,y)\, dA(y) \qquad \text{(visible light scattering)}$$

$$Q(x) = \frac{1}{\pi}\int_S \Big(\epsilon\sigma T(y)^4 + (1-\epsilon)Q(y)\Big) f(x,y)\, dA(y) \qquad \text{(IR scattering)}$$

where:

$$f(x,y) = \frac{[n(x)\cdot(y-x)]_+ [n(y)\cdot(x-y)]_+ V(x,y)}{\pi \|x-y\|^4}$$

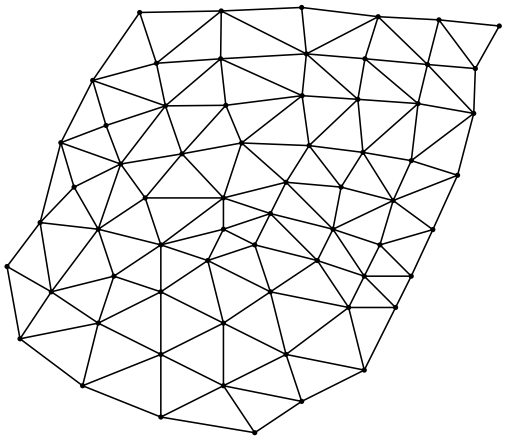
is the "radiosity" kernel.

Challenge is to solve the two integral equations in $O(N)$ time, e.g. fast multipole method not directly applicable because of $V(x,y)$. We developed a fast algorithm for this kernel.

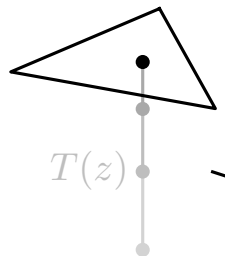
Potter et al., "Precomputed Radiosity for Fast Thermal Modeling on Airless Bodies", *in preparation*.

Precomputed radiosity

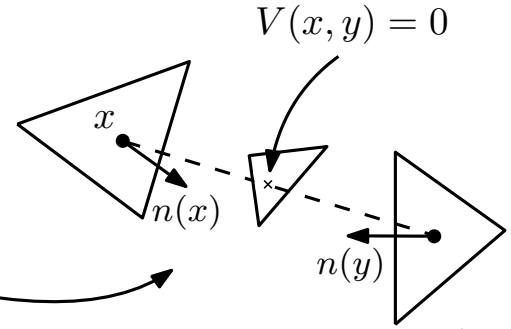
Discretize surface into large tri mesh (easily 3-10M tris)



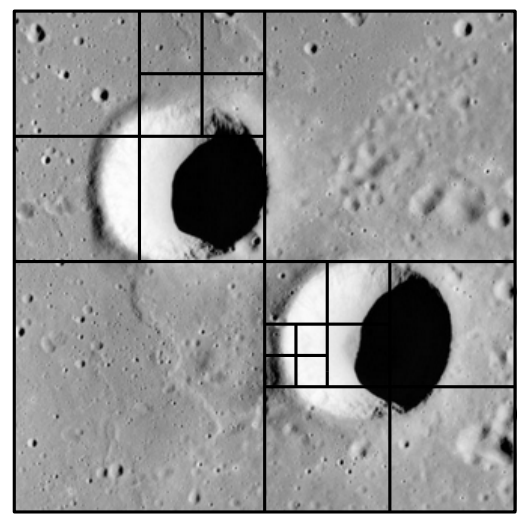
Place 1D thermal model at centroid of each tri



Discretize view factors using midpoint rule



Recursively partition triangles using spatial data structure (quadtree, octree, . . .)



my contribution

assembly

Compress off-diagonal blocks using *either* truncated SVD *or* sparse matrix, depending on which uses less space

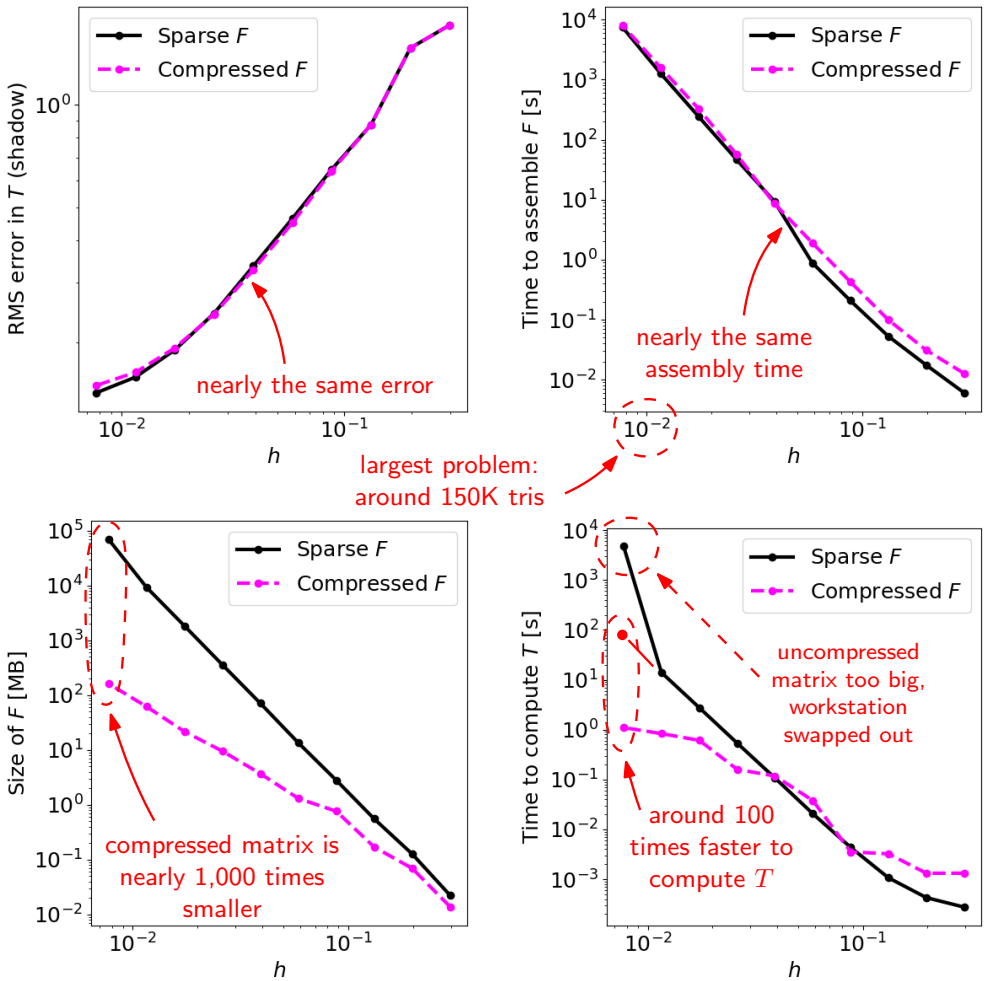
Solve system using simple iterative method (e.g., Jacobi converges in 3–8 iterations)

Drive 1D heat models at each triangle, re-radiated thermal emission for next time step

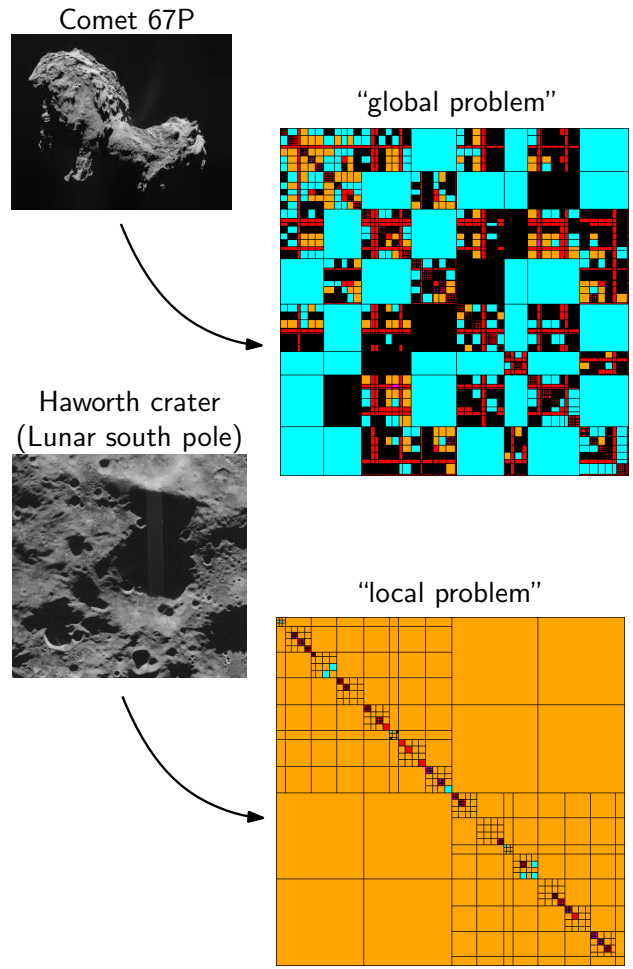
thermal simulation

Numerical results

Validation on hemispherical crater test problem

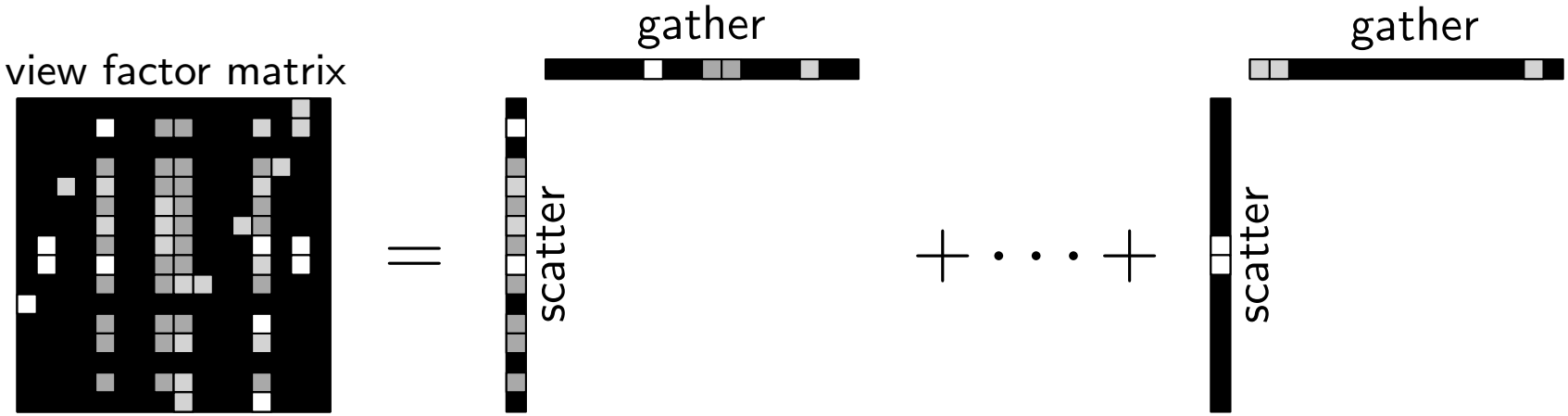


Compressed view factor matrices



E.g.: time step for discretization of Bennu with 3M triangles, takes 3 mins. to compute T: two solves, 8 iterations each, where each MVP takes about 11 seconds.

Future research: nonnegative matrix/tensor factorizations



- Use NMF instead of SVD to compress view factor matrix blocks
- **Compressed sizes around 1.5x smaller for similar accuracy**
- **Terms in NMF are physically meaningful**
- Nonnegative **tensor** factorizations potentially useful for spatial audio codecs and modeling with non-Lambertian BRDFs

References

- Potter, S.F. and Cameron, M.K., 2019. Ordered line integral methods for solving the eikonal equation. *Journal of Scientific Computing, 81*(3), pp.2010-2050.
- Potter, S.F. and Cameron, M.K., 2020. Jet Marching Methods for Solving the Eikonal Equation. *arXiv preprint arXiv:2009.05490*.
- Potter, S.F., Cameron, M.K., and Duraiswami, R., 2021. Precomputed eikonal-based sound propagation. *In preparation.*
- Potter, S.F., Bertone, S., Mazarico, E., and Schörghofer, N. Precomputed Radiosity for Fast Thermal Modeling on Airless Bodies. *In preparation.*

Software: `umiacs.umd.edu/~sfp` and `github.com/sampotter`.

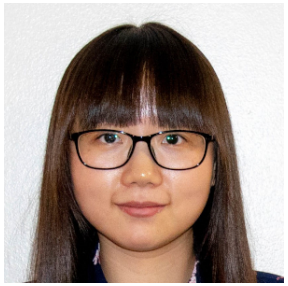
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Masha Cameron
UMD Math



Ramani Duraiswami
UMD CS



Manyuan Tao
UMD AMSC



John Snyder
MSR



Nikunj Raghuvanshi
MSR



Erwan Mazarico
NASA GSFC



Norbert Schörghofer
PSI



Stefano Bertone
NASA GSFC